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Align-Box: Enable Fast Bare-Hand Multi-Object Alignment in VR

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ABSTRACT
In virtual reality (VR, Appendix A: Table A1), efficient alignment of multiple objects is essential for tasks such as indoor layout design, where geometric constraints are common. To improve interaction efficiency and spatial perception, we propose an interactive multi-object alignment metaphor, Align-Box, which employs dual hexahedral boxes and intuitive hand gestures to support both group-based and object-based alignment. By mapping object bounding boxes to proxy boxes, Align-Box enables fast and precise multi-object manipulation. We evaluated the method through a controlled user study. Results show that Align-Box reduced task completion time by up to 77.3% and operation count by 78.6% compared to baseline techniques, while significantly improving usability. These results demonstrate that Align-Box provides an effective workflow for applications such as in-VR scene editing, architectural visualization, and training simulations, and shows strong potential for immersive content creation in Industry 5.0 contexts.

KEYWORDS
Virtual reality; manipulation; multi-object alignment; interaction technique; user study

1. Introduction

Object Alignment is a crucial operation task in multi-object manipulation in virtual reality (VR). Through object alignment operations, users can efficiently group and arrange multiple objects, a task that is indispensable in content creation applications. The goal of object alignment is to achieve spatial constraints on multiple objects through simple operations, typically one or two, that alter their spatial positions. Shi et al. categorized multiple object alignment based on spatial constraints into two types (Shi et al., 2022): one based on aligning objects with external objects outside the target group (object-based alignment) and the other based on aligning a group of objects with itself (group-based alignment). While previous work has experimented with group-based alignment (Shi et al., 2022), it has not provided a general method capable of handling both types of alignment tasks. Specifically, prior techniques like AlignPin (Shi et al., 2022) focus solely on group-based alignment without supporting object-based references, while methods like StickyLine (Ciolfi Felice et al., 2016) are limited by 2D plane constraints. Therefore, this article proposes a general approach for efficiently handling multi-object alignment tasks in virtual reality environments. We position Align-Box as the first unified framework supporting both alignment types within a single workflow.

In VR, users typically interact with objects in the virtual world using two handheld controllers (Mendes et al., 2019; Pfeuffer et al., 2017). These objects often possess six degrees of freedom, aligning them akin to real-world interactions (Mendes et al., 2019; Robinett & Holloway, 1992; Wang et al., 1998). However, interacting with objects in the virtual world using controllers adds extra weight and reduces the immersive experience, which is not ideal for first-time VR headset users.

Furthermore, bare-hand interaction leverages users' natural proprioception and embodied cognition. Unlike controllers, which require mapping intentions to abstract button presses (increasing the cognitive translation cost), bare-hand manipulation allows users to utilize their innate spatial understanding (Kilteni et al., 2012). This theoretical advantage suggests that removing the mediation of a controller can lower the barrier to entry and reduce cognitive load during complex spatial tasks like alignment.

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Fortunately, many modern VR headsets now offer hand-tracking capabilities (Luong et al., 2023; Oudah et al., 2020), making interactions closer to our everyday interactions with physical objects and enhancing users' sense of ownership over their virtual bodies (Adkins et al., 2021). Thus, building upon the proposed multi-object alignment method, this article aims to design a set of bare-hand interactions that enable users to conveniently and effectively perform multi-object alignment in virtual reality scenes.

To achieve these goals, we first define an operational metaphor for multi-object alignment in virtual reality scenes called Align-Box. Align-Box is a cuboid box that can automatically construct itself during the process of selecting objects to be aligned in the scene and supports manual adjustment of size, position, and orientation. Once the box is constructed, an operational proxy of the Align-Box is presented near the front of the user, allowing them to align the selected scene objects within the box through the proxy operation.

Subsequently, we define a set of bare-hand gestures based on the operational process of Align-Box, enabling users to conveniently select and align scene objects rapidly. In Figure 1, from (a) to (d), we show several gestures and features our Align-Box can achieve. Through the featured functions, a user can quickly align scattered objects in order. We conducted a user study involving 36 participants to evaluate and compare the performance of our method with existing alignment methods in terms of alignment efficiency, operational comfort, task load, and usability.

Lastly, we discuss the experimental results, design recommendations, and future research directions.

The main contributions of this work are summarized as follows:

- The design of a virtual reality multi-object alignment metaphor – Align-Box, which utilizes a unique proxy-box mapping to enable combined group- and object-based alignment, supporting efficient multi-object alignment tasks in VR headsets and other immersive environments.
- The design of a minimalist vocabulary of virtual reality bare-hand interaction gestures tailored to Align-Box allows users to carry out multi-object alignment tasks in VR environments comfortably.
- An empirical study involving 36 participants provides design choices and recommendations for the practical use of the proposed Align-Box technology, serving as a comprehensive validation of its effectiveness.

2. Related work

In this section, we briefly review the literature on *Object Alignment* and *Bare-hand Manipulation* techniques in VR. For a detailed understanding of 3D object manipulation, the reader is advised to refer to the survey literature (Mendes et al., 2019).

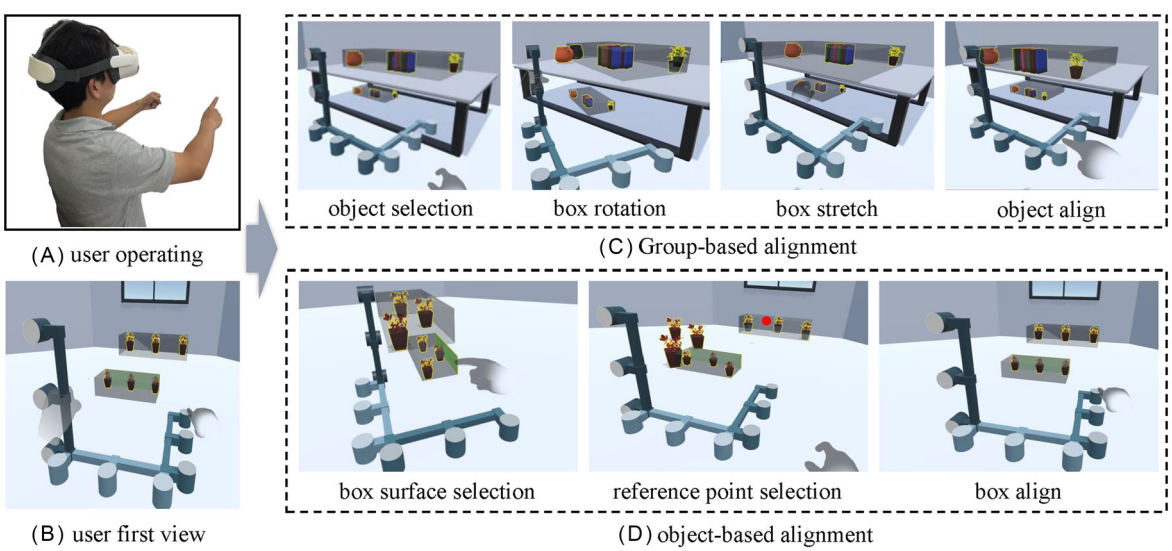

Figure 1. An overview of the Align-Box alignment method. (A) a user wearing a VR headset is using our Align-Box with bare hands. (B) In his view, he can manipulate the virtual objects through the box with several gestures, such as grabbing (left hand) and poking (right hand). (C) through object selection, box rotation, box stretch, and object aligning features, one can conduct group-based alignment to align selected objects to the box's datum in any orientation. Moreover, (D) Align-Box also allows object-based alignment, which uses the box as a medium to achieve alignment of any object's surface by selecting the box surface and any other reference point (red dot) in the virtual scene.

2.1. Object alignment in VR

Object manipulation in VR scenarios is a widely studied topic. Over the past decades, many individual object manipulation methods have been proposed (Frees & Kessler, 2005; Liu et al., 2024; Mendes et al., 2016, 2017; Poupyrev et al., 1996; Wilkes & Bowman, 2008; Zheng et al., 2024). While it is possible to repeat these techniques to manipulate multiple objects, it is a time-consuming and tedious process, especially for tasks that require aligning multiple objects.

Early researchers experimented with the use of constraints for graph dragging, drawing, and alignment in drawing programs (Bier & Stone, 1986; Gleicher, 1992; Raisamo & Räihä, 1996). Later, Uthor et al. (2001) first evaluated 3D manipulation techniques using a 2D device (3-button mouse) and constraints (push away/push against). Stuerzlinger and Smith (2002) described a simple technique for manipulating object groups that aim to increase manipulation efficiency. They introduced two intuitive ways to create a whole group of objects: *drag-add* and *random drag-add*.

After a decade of development, object alignment techniques started to come into the picture for researchers. Some researchers focused on the interaction and alignment of multi-touch interactive displays (Frisch et al., 2011a, 2011b; Wigdor et al., 2011). Frisch et al. (2011a) proposed two tools-interactive grids and multi-touch alignment guides-which support layout tasks on interactive display. They can be used to arrange objects along straight lines and curves. Then, they subsequently introduced *Neat* (natural and effective layout techniques), which provides a consistent set of multi-touch tools and gestures for aligning and distributing graphical objects on interactive surfaces (Frisch et al., 2011b). Wigdor et al. (2011) introduced *Rock & Rails*, a set of interactions with shape gestures to enable precise spatial manipulations. Xu et al. (2015) proposed GACA, which enables the arrangement of graphic elements in groups. The key idea of the technique is an issued arrangement command that instructs how to automatically decompose a 2D layout into multiple 1D groups.

Later, with the development of virtual reality/augmented reality head-mounted displays and gesture recognition and tracking technologies, research on multi-object alignment gradually transitions into immersive virtual reality environments (Barrera-Machuca et al., 2020; Ciolfi Felice et al., 2016; Hayatpur et al., 2019; Shi et al., 2022; Song et al., 2012; Wolfartsberger et al., 2017). Song et al. proposed a handlebar metaphor to manipulate a single virtual object with mid-air interaction while supporting packing/aligning multiple objects along a line (Song et al., 2012). Ciolfi Felice et al. (2016) created *StickyLine*, a tool that can create precise, predictable, and persistent interactive alignment and distribution relationships for graphical objects. Recently, Hayatpur et al. proposed *Plane, Ray, and Point*, a set of interaction techniques that utilizes shape constraints to enable quick and precise object alignment and manipulation in virtual reality (Hayatpur et al., 2019). However, the alignment only works for a single object. Then, Shi et al. technically defined multi-object alignment in VR and introduced four techniques: *AlignPanel, AlignWidget, AlignPin, and AlignGesture*. They also conducted a user study to evaluate their performance in VR scenarios (Shi et al., 2022).

In summary, multi-object alignment is still an under-explored field, and there is an urgent need to investigate a more complete and robust interaction alignment method. Therefore, this article proposes the Align-Box technique, which aims to form a complete set of efficient alignment processes for objects in immersive virtual scenes.

To better situate our work, Table 1 provides a systematic comparison of Align-Box with several key existing techniques, including *StickyLine* (Ciolfi Felice et al., 2016), *Plane, Ray, and Point* (Hayatpur et al., 2019), *AlignPin* (Shi et al., 2022), and the *WIM* (Stoakley et al., 1995).

Furthermore, recent advancements in immersive technologies have highlighted the importance of spatial alignment in industrial contexts. Research on AR/VR-assisted assembly demonstrates that efficient object

Table 1. Comparison of Align-Box with related techniques including gesture complexity and device requirements.

Technique	Interaction Modality	Primary goal	Group-based Alignment	Object-based Alignment	Arbitrary Orientation	Gesture Complexity	Interaction Precision	Device Req.
StickyLine	Multi-touch/stylus	2D layout	Yes	Limited	No	Medium	High	Touch surface
Plane, Ray, Point	Controller	Single object precision	No	Yes	Yes	Low	High	Controller
AlignPin	Controller/hand	Group alignment	Yes	No	No	Low	Medium	Controller/hand
WIM	Controller	General manipulation	Manual	Manual	Yes	Medium	Low	Controller
Align-Box (Ours)	Bare-hand	Multi-object alignment	Yes	Yes	Yes	Low (4 poses)	High (Proxy)	VR headset

positioning is critical for autonomous assembly systems and digital twin interactions (Bahubalendruni et al., 2015; Eswaran et al., 2023, 2024; Eswaran & Bahubalendruni, 2024). These works suggest that optimizing the alignment workflow can significantly enhance operator performance in broader Industry 5.0 scenarios.

2.2. Bare-hand interaction

Hand-tracking technology has evolved dramatically over the past decade, and bare-hand interactions are gradually replacing controller-based interactions. Comparative experiments have been conducted to compare the new bare-hand interactions with the traditional controller-based interactions (Fahmi et al., 2020; Khundam et al., 2021; Luong et al., 2023; Rantamaa et al., 2023). Despite the advantages of each, natural bare-hand interaction without adding extra weight is considered the dominant interaction method for the future of immersive virtual reality.

Advances in hand-tracking technology have catalyzed bare-hand interaction in virtual reality (Battiato et al., 2018; McMahan et al., 2010; Oudah et al., 2020; Tong et al., 2023). Many researchers have evaluated the performance of real bare hands when performing virtual reality interactions (Adkins et al., 2021; Kangas et al., 2022; Li et al., 2018; Lin et al., 2019; Navarro & Sundstedt, 2019). Li et al. (2018) evaluated the effects of four virtual cursor offset techniques on 3D object selection performance with hand-tracking interaction. Navarro et al. investigated how the user experience and task efficiency are influenced by variations between the size of a user's actual hand and their avatar's hand Navarro and Sundstedt (2019). Adkins et al. (2021) tried to study the grasp visualization's influence on ownership, realism, efficiency, enjoyment, and presence.

In addition to investigating the impact of bare-hand interaction on virtual reality applications, many researchers are also focusing on developing new bare-hand interaction methods in virtual environments (Han et al., 2024; Normand et al., 2024; Pei et al., 2022; Shi et al., 2023). Pei et al. (2022) proposed an interaction technique that lets users' hands become virtual objects by imitating the objects themselves. This interesting interaction is creative but limited in the number of gestures and detection accuracy. Shi et al. (2023) proposed the multiFingerBubble technique, which enables multi-object selection with the tracked fingers. Recently, Han et al. (2024) introduced the VR-Handnet approach, which can produce physically plausible hand motion when interacting with virtual objects. Normand et al. (2024) investigate the effect of visual hand rendering and haptic hand rendering on the bare-hand interaction.

Bare-hand interaction is gradually replacing traditional handheld controllers as the mainstream interaction method for next-generation immersive virtual reality environments. Therefore, this article aims to design a bare-hand interaction scheme suitable for the alignment method in this article.

3. Method

This section describes the details of our Align-Box method, which aims to form a complete set of efficient multi-object alignment processes in immersive virtual scenes. We first introduced the bare-hand object alignment design principles in subsection 3.1. Then, we outline the basic design of our Align-Box in subsection 3.2 and subsequently introduce how to interact with it in subsection 3.4. Finally, we define a set of hand gesture interactions involving four gestures to conduct object alignment with our Align-Box in subsection 3.4.

3.1. Design principles

We aim to develop an efficient, comfortable, and easy-to-use method of aligning objects with bare hands. To this end, we considered four design principles to ensure this. These principles are as follows:

Near-field Interaction: Near-field interaction emphasizes placing interactable objects and triggers within the user's natural reach. This principle is widely adopted in many virtual reality systems, as it is intuitive, natural, and facilitates ease of interaction (Montano Murillo et al., 2017; Wu et al., 2024). It allows users to operate without large body movements or overextension of the arms.

Usability: The designed solution should utilize commonly used poses accumulated in the current VR gesture interaction practices instead of complex poses. Commonly used poses are easy to learn and use in actual operations, conform to user intuition, and distinguish different operations through gestures.

Error Tolerance: The designed solution should be able to identify and process non-standard gestures correctly. Secondly, to reduce misoperations, the method should set up a "confirmation" mechanism for operations that are easy to mistouch, requiring users to confirm their intentions before performing key operations.

3.2. Align-Box

Following the three principles mentioned above, we define the Align-Box as a pair of hexahedral enclosing boxes. As shown in Figure 2, one box is the *Alignment Box*, which encloses the target object to be aligned in the virtual scene. The other is the *Proxy Box*, a manipulable proxy presented to the user. The *Proxy Box* features an operable *Alignment Axis* and *Pin Button*, and user interactions with the *Proxy Box* are reflected to the *Alignment Box* to achieve object alignment in the scene.

The *Proxy Box* includes three interactive elements: the Pin Button, the *Alignment Axis*, and its six faces. Users can poke the Pin Button to align the selected object. Similar to AlignPin in previous literature (Shi et al., 2022), each *Alignment Axis* has three *Pin Buttons* for center and edge alignment. Users can adjust the orientation of the Align-Box by grabbing the *Alignment Axis* to rotate the *Proxy Box*. The six faces of the *Proxy Box* can be grabbed and dragged to adjust the size of the *Alignment Box*. Users can also poke a face to select it and align it with a reference point in the scene for quick position adjustment of the *Alignment Box*. Align-Box can achieve both group-based alignment and object-based alignment. Group-based alignment can be directly implemented through the Align-Box, while object-based alignment can be achieved by aligning the Align-Box with the reference object. The details will be introduced in subsection 3.3.

To ensure user accessibility, the Align-Box method defines the *Proxy Box* interaction area as a $60cm \times 60cm \times 60cm$ cube in front of the user's viewpoint, centered 50 cm in front of the viewpoint and 20 cm below. Generally, the length of the *Alignment Axis* matches the sides of the *Proxy Box*, and the scale ratio between the *Proxy Box* and the *Alignment Box* is fixed at 0.1. If the size of *Alignment Box* is too small after selecting an object, causing the longest side of the *Proxy Box* to be less than the minimum operable length (set at 15 cm), the *Proxy Box* and Pin Button will be too crowded. Therefore, we set the minimum length and the maximum length of the *Alignment Axis*. The scaling of the *Proxy Box* is adjusted so that its longest side ranges from 15 cm to 60 cm.

For the implementation, we defined specific thresholds for gesture recognition. A *Poke* gesture was registered when an index fingertip's velocity exceeded 0.2 m/s toward an interactive element and came within a 2 cm proximity. A *Grab* was recognized when the distance between the thumb and index fingertips was less than 1.5 cm. To ensure direct and responsive control, we did not apply any smoothing to the raw hand tracking data. The fixed 0.1 scale factor for the *Proxy Box* was chosen based on pilot

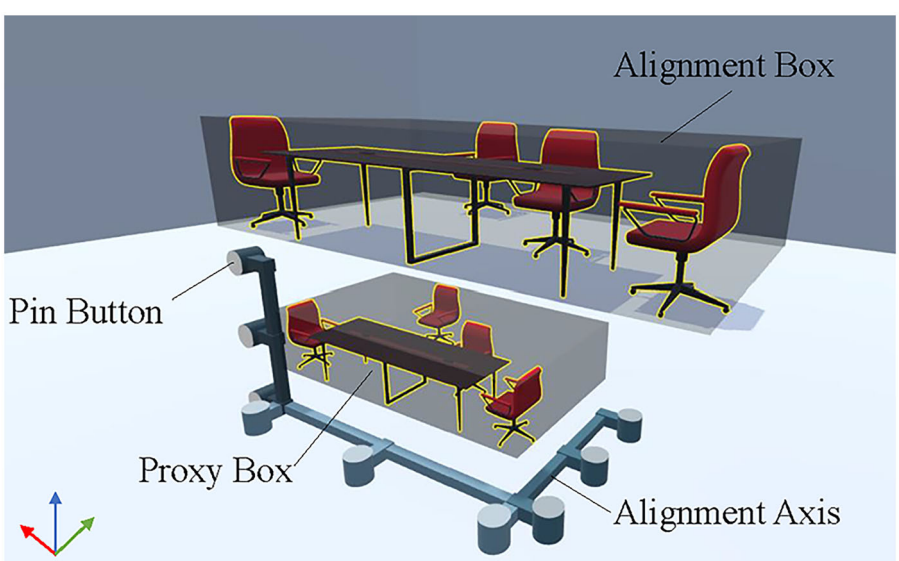

Figure 2. A basic construction of the Align-Box, which consists of the *Alignment Box* and the *Proxy Box*. Around the *Proxy Box*, there are *Alignment Axes* and *Pin Buttons* for users to apply their customized alignment.

studies, as this ratio was found to offer a comfortable balance: it makes distant or large alignment boxes manageable in the near-field interaction space without making the proxy so small that fine-grained adjustments become difficult. Similarly, the gesture recognition thresholds were calibrated through an iterative process with 5 pilot users to minimize false positives while maintaining responsiveness.

3.3. Interaction design

Align-Box relies on bare-hand interactions on the *Proxy Box*, *Alignment Axis*, and *Pin Button* to align selected objects in the *Alignment Box* (Figure 2). This section presents each of these interactions to describe the Align-Box operation mechanism. Appendix B shows detailed geometric calculations.

3.3.1. Object select

Before conducting object alignment, there must be a multi-object selection process. It is convenient to select multiple objects for Align-Box. The hand cursor triggers the selection. After the user selects the first object, the *Alignment Box* is constructed, the same size as the object's axis-aligned bounding box. The *Proxy Box* is computed using the *Alignment Box* of the object. Whenever the user selects a new object outside the current *Alignment Box*, the system immediately expands the *Alignment Box* to enclose the new object. Then it updates the *Proxy Box*. To assist multi-selection, the wrapped unselected objects will also be set as selected when the *Alignment Box* expands. This enables quick selection for a bunch of objects spatially adjacent. If the user wishes to exclude certain selected objects, they can be dropped by reselecting them with the hand cursor.

Figure 3 shows a fast object selection process. In Figure 3(A), the user applies two cursor selections, one for the books and the other for the right-most plants. Only the enclosed books and right-most plants are selected. Then, after selecting the left-most orange cup (in Figure 3(B)), the *Alignment Box* is extended, and the wrapped three other plants are also selected. Meanwhile, the *Proxy Box* keeps updating during selection.

Accompanying the assisted multi-selection is the quick deselecting of all objects. In practical alignment scenarios, deselecting all objects after aligning is an inevitable and tedious task. To improve efficiency, this method designs a gesture trigger: users can make a thumbs-up gesture to trigger the "deselect all" function, thereby canceling the selection status of all selected objects. This can improve the speed and convenience of completing successive alignment tasks.

3.3.2. Object align

Align-Box allows users to poke the *Pin Buttons* on the *Alignment Axes* with a poking gesture to apply group-based alignment. This design is similar to that of the AlignPin (Shi et al., 2022).

The alignment methods include center alignment and edge alignment. Center alignment makes the object's center point coincide with the center plane of the *Alignment Box*, while edge alignment makes

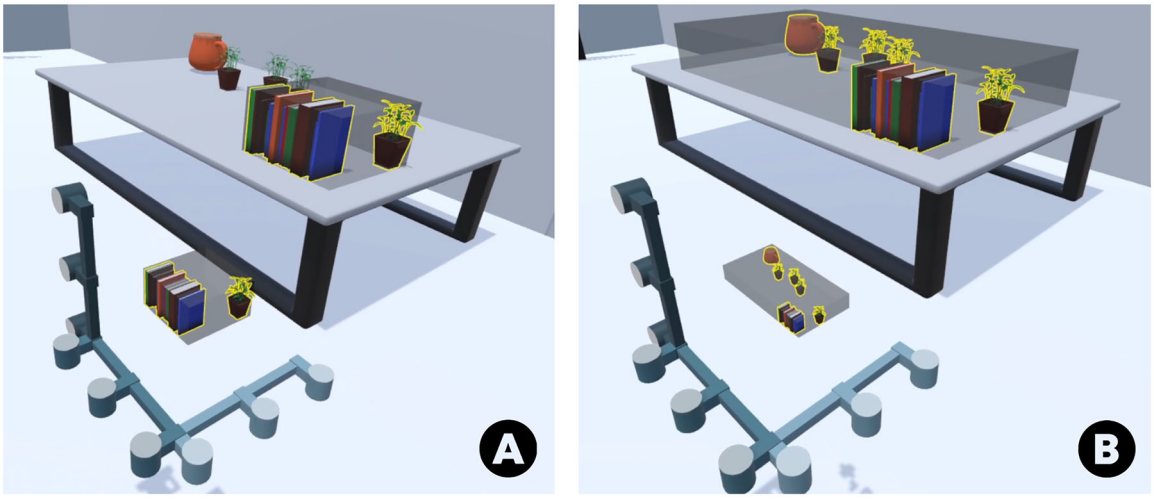

Figure 3. In the state shown in (A), select the outermost cup to extend the selection range to cover a total of 6 objects in (B).

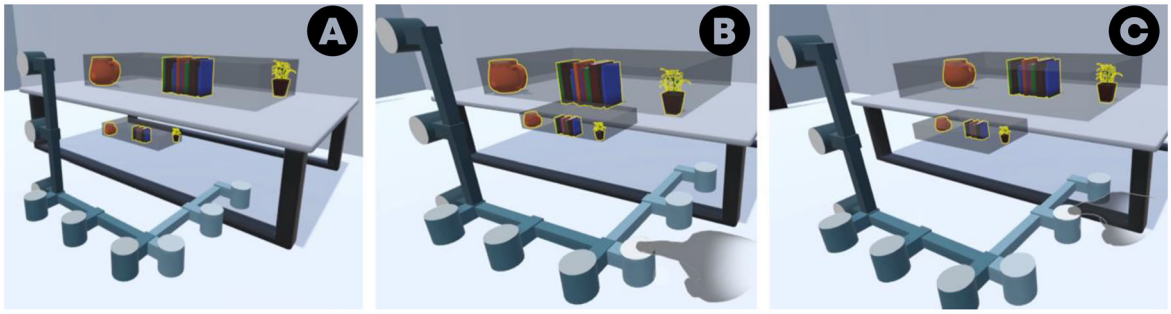

Figure 4. (A) Scattered objects on a table. (B) Poking the edge button applies edge alignment, aligning the selected objects along the side surface of the *Alignment Box*. (C) Poking the middle button applies center alignment, aligning the selected objects on the center plane of the *Alignment Box*.

the object's edge coincide with the side surfaces. There are 3 pins on each Alignment Axis in each dimension. The middle pin is used for center alignment, and the pins at both ends are used for edge alignment. As shown in Figure 4, scattered objects are on the table (A). (B) shows the result of the edge alignment when the user pokes the button at the edge of the Alignment Axis. (C) shows the result of the center alignment when the user pokes the middle button of the Alignment axis.

Regarding usability, each *Pin Button* is designed to be pressable on both sides to ensure the buttons remain in a comfortable interaction area after rotation. This design also allows users to poke the *Pin Button* from different positions. In terms of fault tolerance, to reduce the likelihood of misoperation, users need to poke the *Pin Button* twice to complete the alignment. After the first poke on the *Pin Button*, translucent proxies of the selected objects will appear in the *Alignment Box*. Their positions reflect the effect of the alignment. After the second poke, the selected objects will be aligned.

3.3.3. Box stretch

In many cases, the surfaces of the axis-aligned bounding box of selected objects are not in the desired alignment position. Therefore, we provide the Box Stretching feature, which allows the user to adjust the surface's position by grabbing the surface of the *Proxy Box* and dragging it. The grabbed face can be stretched outward or shrunk inward while the rotation angle and the positions of the other faces remain unchanged. By stretching the *Proxy Box*, the *Alignment Box* is also synchronously changed.

Since the *Alignment Box* works as the alignment reference during *Object Align*, changing its scale in different directions through box stretching enables more flexible alignment. For example, users can align objects to a surface outside the current *Alignment Box*. As shown in Figure 5, for the scattered objects on the table (A), users can first grab the *Proxy Box*'s front surface, drag to align the *Alignment Box*'s front surface to the edge of table (B), and then poke the *Pin Button* on the *Alignment Axis* to align the scattered objects along the table edge (C).

3.3.4. Box rotate

In addition to the box stretching feature, we also provide a box rotation feature to allow the user to adjust the orientation of the *Alignment Box*. Grabbing the *Alignment Axis* triggers the box rotation function.

The box's rotation function is triggered by grabbing the alignment axes. The Align-Box shares three alignment axes: the x-axis, y-axis, and z-axis. When the system detects that the user has grabbed one of the *Alignment Axes*, it records the position of the hand's grabbing point. Subsequently, the user can move their hand to control the rotation angle. Both the *Proxy Box* and the *Alignment Box* rotate around an axis that passes through their centers and is parallel to the currently active alignment axis.

Figure 6 shows a practice using of box rotation feature. For the scattered objects (A), the user first grabs the vertical *Alignment Axis* and rotates the boxes horizontally to the desired orientation (B). Then, the user pokes the *Pin Button* to complete a center alignment (C). By rotating the box, users can align objects in an oblique direction and arrange objects in a non-orthogonal direction.

During the rotation process, a translucent preview of the selected objects will be visualized in the *Alignment Box* to enhance usability. This preview will rotate along with the *Proxy Box* and the

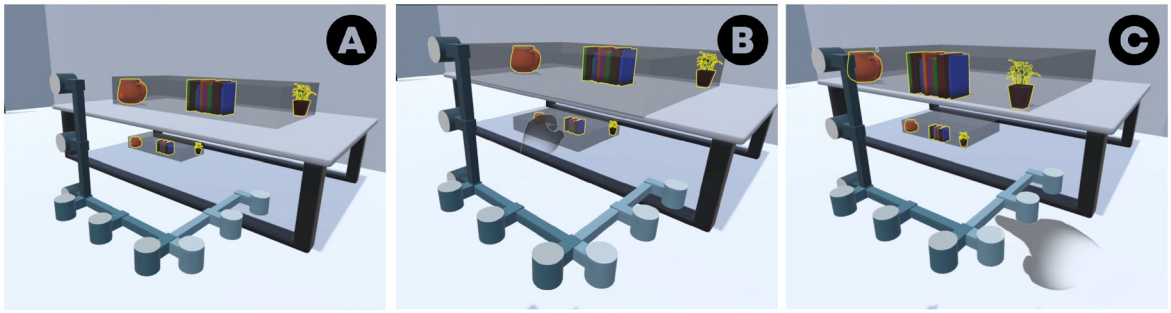

Figure 5. (A) Scattered objects on a table. (B) The user is grabbing and dragging the front face of the *Proxy Box* to align that of the *Alignment Box* to the table's front edge. (C) By poking the *Pin Button* on the *Alignment Axis*, the objects are aligned along the table's edge.

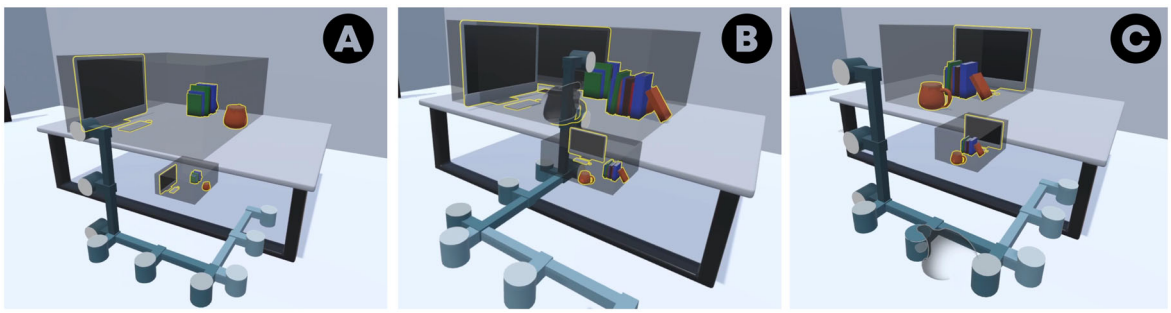

Figure 6. (A) Object's initial state. (B) Grab the vertical Alignment Axis to rotate the box horizontally. The translucent preview of the selected objects and *Alignment Box* will be generated. (C) Poke the Pin Button to complete the alignment.

Alignment Box according to the user's actions, intuitively displaying the outcome of the operation. In terms of fault tolerance, before the user presses the *Pin Button* to finalize the alignment, any box rotation will not change the position or orientation of the selected object. The operation results remain in a preview state, reducing the possibility of errors. Furthermore, the user can cancel the current rotation operation at any time, and all objects will reset to their state before the current rotation operation.

3.3.5. Box align

Above, we discussed the alignment processes and interactions inside the Align-Box, which enables a flexible group-based alignment. To facilitate object-based alignment, we introduce box-aligning interactions.

The box aligning process consists of two main steps: select a specific surface of the *Proxy Box* and select the desired alignment point in the virtual environment. The aligning effect is that the pose of the *Alignment Box* will be changed to make the center point of the selected *Alignment Box*'s surface coincide with the desired point and the inner normal of the center point equal to the normal of the desired point.

Figure 7 shows a basic process of box aligning. For the three plants selected in the virtual scene (A), the user first pokes on the right surface of the *Proxy Box*, and the face will be highlighted for clarity (B). Then, the user clicks on a point on the wall with a cursor. The system automatically calculates the pose of the *Alignment Box* and generates a translucent preview (C). Finally, the user pokes the *Pin Button* and completes the alignment process (D). Users can conveniently align the selected objects to an oblique position and orientation in the virtual environment through the box aligning process.

Similar to the box rotate feature, the highlight and preview visualization effects are used to prompt users about the current state, enhancing usability. Meanwhile, the final alignment of the selected objects only happens after poking the *Pin Button* to reduce misoperations. To distinguish it from the Stretch Box operation, we use the poking gesture to trigger the box-aligning operation.

A significant part of aligning multiple objects involves existing planes in the scene, such as aligning a painting to a wall or a chair to the edge of a table. These planes act as the baseline for alignment but are not the objects being aligned. Using the Box Stretch and Box Rotate features to adjust the

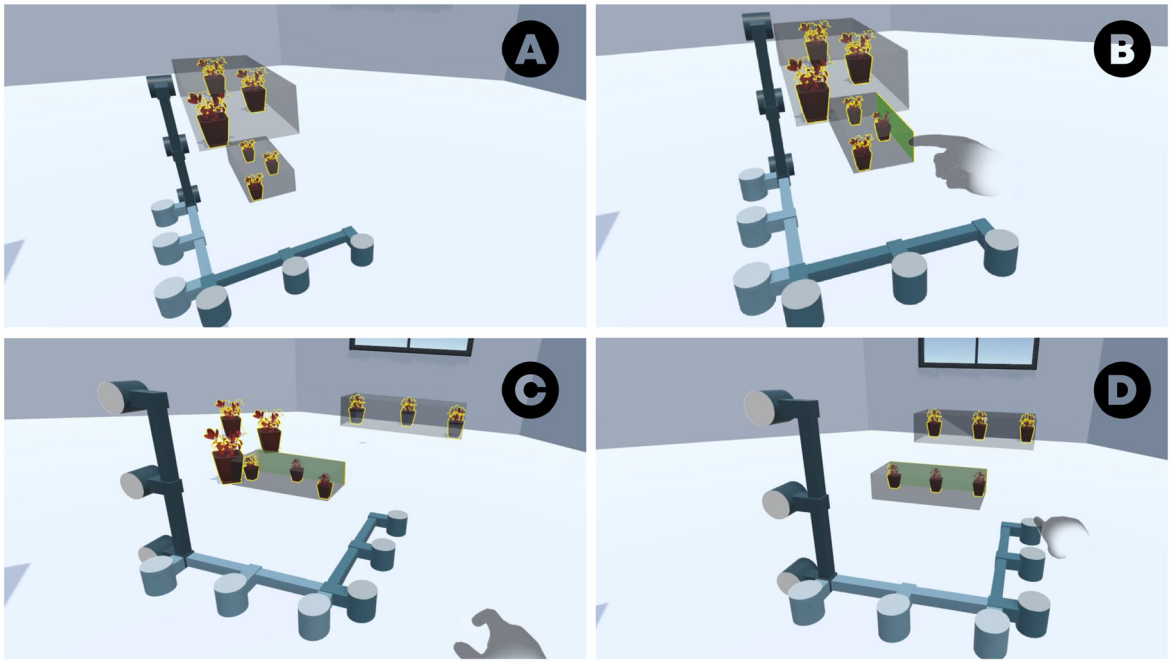

Figure 7. (A) Initial state. (B) Poke on the right face of *Proxy Box*, and the face will be highlighted. (C) Click on the point on the wall, and the translucent preview of *Alignment Box* will be generated. (D) Poke the Pin Button to complete the alignment of the objects against the wall.

Alignment Box to the target alignment position requires complex and precise operations, which can easily increase the user's workload. With the Box Align feature, users can quickly define the 6-DoF position of the *Alignment Box* using environmental planes, enabling efficient alignment of multiple objects with non-orthogonal surfaces, i.e., object-based group alignment.

To summarize the core workflow of Align-Box, we present the interaction logic in Algorithm 1. The algorithm outlines the main event loop, handling user inputs for object selection, box manipulation, and the final alignment command. Each step is designed to provide clear feedback and maintain a stateful, predictable interaction.

Algorithm 1. Align-Box Core Interaction Logic

Input: User's hand gestures, virtual scene objects
Initialize *SelectedObjects* list as empty; *AlignmentBox* as null
loop
  Detect user gesture *G* and target component *C*
  if *G* is *Cursor* on an object **then**
    Toggle selection of object in *SelectedObjects*
    Update *AlignmentBox* to bound all *SelectedObjects*
    Update *ProxyBox* based on *AlignmentBox*
  else if *G* is *Grab* on a *ProxyBox* component (face/axis) **then**
    Stretch or Rotate *ProxyBox* and *AlignmentBox*
  else if *G* is *Poke* on a *ProxyBox* face **then**
    Mark face for Box Align; await cursor input for target
  else if *G* is *Poke* on a *PinButton* **then**
    Preview alignment of *SelectedObjects*
    if *C* is poked a second time **then**
      Finalize alignment of *SelectedObjects*
    end if
  end if
end loop

Table 2. Gestures (images from left to right: *Cursor*, *Poke*, *Grab* and *Confirm*) used in Align-Box and their functions.

Components \ poses				
Proxy Box		Box Align	Box Stretch	Deselect All
Pin Button		Object Align		
Alignment Axis			Box Rotate	
Objects	Object Select/Box Align			

3.4. Gesture design

To effectively use the interactive features of the Align-Box method for multi-object alignment, we design a set of corresponding gestures for interaction. Our design follows two principles:

- The number of gestures involved in the interaction should be minimized to reduce the user's learning burden.
- The gestures should be simple and easy to distinguish to avoid misrecognition, enhancing usability and fault tolerance.

Align-Box uses four gestures: two specialized gestures (*Grab* and *Poke*) for direct interaction with Align-Box, and two general gestures (*Cursor* and *Confirm*) related to a specific VR platform. Table 2 summarizes these four gestures and their functions within Align-Box.

The general gestures are primarily for selecting/deselecting interactions. We use the *Cursor* and *Confirm* gestures in our context (depending on the VR platform). The *Cursor* gesture corresponds to two steps in the Align-Box method: selecting objects when initially constructing the Align-Box and selecting alignment points in the virtual scene for the Box Align feature. The presence of a highlighted face in the *Proxy Box* (Figure 7(C)) distinguishes these steps, preventing interference. The *Confirm* gesture corresponds to the deselecting step. When this gesture is recognized, all selected objects are deselected from the selection state, and the constructed Align-Box is destroyed.

We selected two classic hand gestures for the specialized gestures. Interaction functions are defined by combining gestures with target components. As shown in Table 2, the *Poke* gesture is primarily responsible for performing align functions, for both *Box Align* and *Object Align*, and the *Grab* gesture is responsible for adjusting the boxes, including Box Stretch and Box Rotate features.

The following features of the gesture design of the Align-Box method avoid misrecognition and ensure the system's usability. Firstly, in the virtual environment, the *Proxy Box* is spatially separated from the *Pin Button* and *Alignment Axis*, avoiding misrecognition from the same gesture (*Poke* and *Grab*). Secondly, for the spatially adjacent *Alignment Axis* and *Pin Button*, the gestures paired with them are the *Grab* gesture and the *Poke* gesture, respectively. The significant difference between the two gestures helps to prevent the misrecognition of functions. Finally, among the four gestures utilized, the *Cursor* and *Confirm* gestures are general and vary depending on the VR platform, while Align-Box predominantly employs two specific gestures (*Grab* and *Poke*), ensuring simplicity in its operation.

In summary, our Align-Box method employs as few specialized gestures as possible, complemented by the general selection/deselection gesture of VR devices. This defines a simple gesture set suitable for the method, making it easy to use and enhancing its compatibility. We also refer the reader to the accompanying video showing the method in action.

It is worth noting that the final selection of these four gestures was determined after an iterative pilot study involving 5 expert users. This process ensured that the chosen gestures were ergonomically comfortable and distinct enough to minimize recognition errors during rapid operation.

3.5. System performance

To ensure the reliability and usability of Align-Box, we evaluated two key performance aspects: gesture recognition accuracy and overall system latency.

Table 3. Confusion matrix for gesture recognition.

Actual \ predicted	Cursor (%)	Poke (%)	Grab (%)	Confirm (%)	None (%)
Cursor	97	1	0	1	1
Poke	1	96	2	0	1
Grab	0	1	98	0	1
Confirm	1	0	0	94	5

3.5.1. Gesture recognition performance

Our system relies on four distinct hand gestures as described in Section 3.4. We conducted a performance test to evaluate the recognition accuracy using the Meta Quest 2's Hand Tracking 2.0 API. We collected 500 samples for each gesture from 10 users (not participants in the main study). The overall recognition accuracy was 96.2%. The detailed performance is presented in the confusion matrix in Table 3.

The results show high accuracy for all gestures, with minimal confusion between the primary interaction gestures (*Poke* and *Grab*). The main source of false positives was misclassifying a neutral hand state as a specific gesture, which was minimized by requiring clear and deliberate hand poses.

3.5.2. System latency

We measured the end-to-end system latency, defined as the time from the user's hand completing a gesture to the corresponding visual feedback (e.g., object alignment preview) appearing on the display. The average system latency was measured to be 48.3 ± 5.1 ms. This low latency ensures that the interaction feels responsive and instantaneous to the user, which is critical for maintaining immersion and usability in VR.

4. User study

4.1. Overview

We conducted an empirical user study to assess the performance of our Align-Box method. Our comparison included two prior methods: AlignPin (Shi et al., 2022), which focuses on alignment tasks, and WIM (Stoakley et al., 1995), designed for object manipulation within indoor scenes. Align-Box was specifically compared with the WIM method due to similarities in their proxy manipulation mechanisms. The primary objective of the study was to determine if the Align-Box method effectively accomplishes general alignment tasks in virtual environments.

Accordingly, we formulated the following hypotheses:

- **H1**. Align-Box can significantly improve the efficiency of completing the alignment tasks.
- **H2**. Align-Box can provide users better user experience.

4.2. Participants and apparatus

We recruited 36 participants (20 males and 16 females), all with normal vision (or corrected to normal vision with glasses), no balance disorders, right-handed, and voluntarily participated in this user study. Among them, 28 users have experience operating virtual reality headsets.

The experimental equipment is the Oculus Quest 2, a classic virtual reality headset that can perform bare-hand recognition and gesture recognition. The headset has a display resolution of 1832×1920 per eye, 89° horizontal field of view, and a refresh rate of 120 Hz. The headset supports Meta's Hand Tracking 2.0, introduced in 2022, which applies a new method based on deep learning to optimize the continuity of hand tracking and better predict and recognize hand occlusions and fast movements, thereby minimizing difficulties or errors in hand tracking that could interfere with the experiment or affect the evaluation of the method. To control for the potential impact of prior familiarity, participants were stratified by experience level and evenly distributed across the Latin Square sequences to ensure a balanced allocation. A post-hoc power analysis confirmed that our sample size of $N = 36$ provided sufficient statistical power (>0.8) to detect the observed effect sizes at a significance level of $\alpha = 0.05$, validating the reliability of our results.

The headset was connected to a Windows 11-based laptop with an Intel i7-12650H CPU @ 2.30 GHz, an NVIDIA GeForce RTX 4060 Laptop GPU, and 16GB of RAM. The tasks were performed in a room familiar to the participants, with all participants completing the tasks in a standing position.

Before the experiment, each participant underwent a brief calibration and setup procedure. This included adjusting the headset's physical interpupillary distance (IPD) slider to match their vision and tightening the straps for a secure fit. The study was conducted in a well-lit indoor environment with ample space to minimize occlusion from physical objects. While no formal software calibration was required by the Quest 2 system, these manual adjustments are crucial for user comfort and perceptual accuracy. The reliance on user-specific IPD settings and a controlled lighting environment means that generalizability of the findings might be affected in non-ideal settings.

4.3. Study design

There were three conditions in our study. For control condition 1 (CC_1), participants used the *AlignPin* method. The original article focuses exclusively on the alignment mechanism and does not specify a method for multi-object selection. To create a fully functional baseline for our end-to-end task, we implemented a standard ray-casting selection technique for the AlignPin condition. In this implementation, users pointed at objects with a ray emanating from their hand and performed a pinch gesture to select or deselect them. We chose ray-casting because it is a conventional and efficient selection method in VR and is not a novel contribution of our own technique. This design choice ensures a fair comparison that focuses squarely on the alignment mechanics of Align-Box versus AlignPin, rather than being confounded by the selection process. For control condition 2 (CC_2), they used the *WIM* method (Stoakley et al., 1995). For the experimental condition (*EC*), they used our Align-Box method. We selected these two baselines to represent the state of the art in two relevant areas: *AlignPin* as a recent, specialized multi-object alignment technique, and *WIM* as a classic and widely recognized general manipulation technique. This comparison allows us to evaluate Align-Box against both specialized and general-purpose interaction paradigms.

The alignment tasks followed these standards:

- *Diversity*: The tasks in the user study should cover as many alignment scenarios as possible, including different quantities, types, alignment difficulties, and object sizes.
- *Intuitiveness*: The target positions for all objects that need to be aligned should be self-explanatory, and users can clearly understand the expected outcome without additional explanations.
- *Practicality*: The task design should be close to the actual application and needs to genuinely reflect the effectiveness of alignment methods in practical scenarios.

Based on these standards, the user study was designed to occur in an office scene as shown in Figure 8(A). Figure 8(B) shows the scene before alignment.

Since the AlignPin method is unable to perform object-based alignment and rotating tasks. We split the user study into two phases. For CC_1, participants were only asked to accomplish tasks in phase $P1$. There are a total of four tasks, with one task in the first phase and three tasks in the second phase. In phase $P1$, the user should align the coffee pots on the table in a row. In phase $P2$, the user should align books and bookshelves to the wall, rotate a computer to face inward and align flower pots to a non-orthogonal corner (as shown in Figure 8(C)). These tasks correspond to different quantities, distances, object sizes, and types of alignment. Details are shown in Table 4. These tasks were specifically chosen to simulate real-world scenarios in professional domains. For instance, aligning books and shelves represents a typical interior design workflow, while rotating monitors and arranging plants mimics the layout adjustments required in digital twin scene editing and architectural visualization.

4.4. Procedure and metrics

The user study was conducted in accordance with ethical guidelines and received approval from our institution's Institutional Review Board (IRB). Users followed the procedure illustrated in Figure 9 to

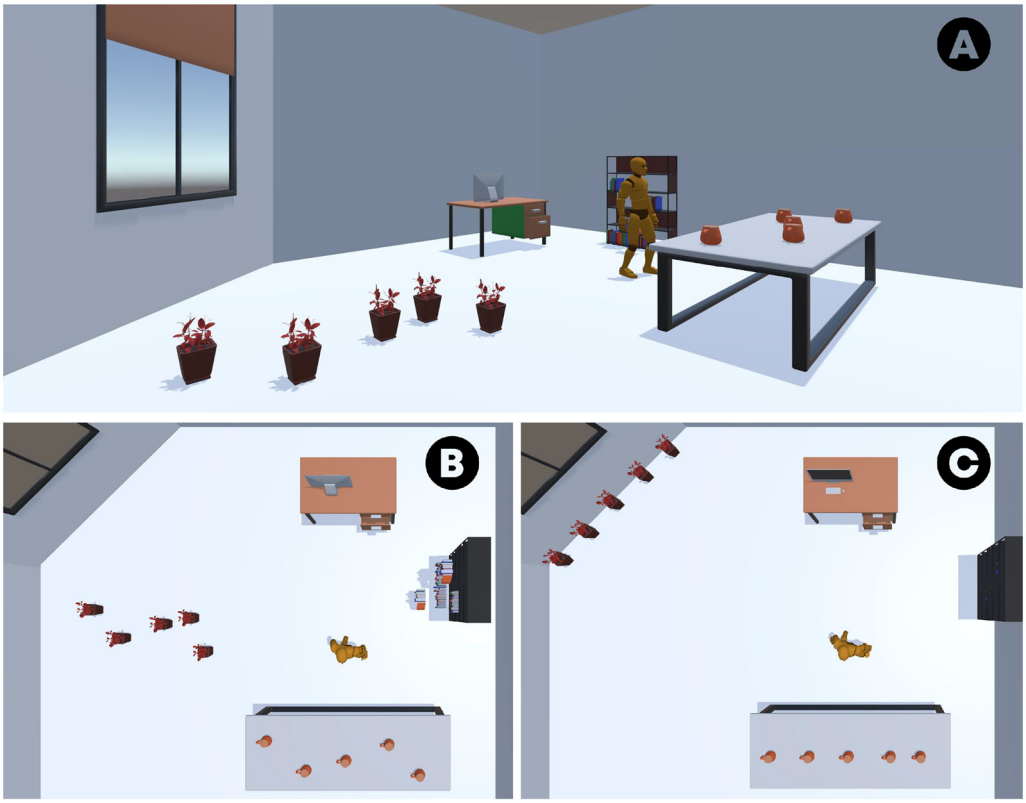

Figure 8. Front (A) and top (B) view of the user study scene before alignment. (C) Expected alignment outcome. The yellow avatar marks the position of the participants when they are conducting the alignment tasks.

Table 4. Detailed information for four tasks in our user study.

Phase	Objects	Distance	Object Count	Alignment Type	Capacity CC_1	Capacity CC_2	Capacity EC
P1	Pots	close	5	Group-based	√	√	√
P2	Shelf	Medium	5	Align to Orthogonal Surface		√	√
	PC	Medium	1	Rotate Object		√	√
	Plants	Far	5	Align to Non-Orthogonal Surface		√	√

complete the user study. In the preparation stage, they were first informed of the study's purpose and tasks (to accomplish the alignment tasks as fast as possible). After five minutes of practicing the operations for the three conditions. The balanced Latin square determined the order of the conditions. Then, the main study started, and the participants completed the alignment tasks and filled out the task load and the usability questionnaires for each condition. There was a two-minute break before changing conditions. During the main study, their behavioral data were recorded for further analysis. Once the main study was concluded, the experimenter would engage in simple interviews with the participants to gather their perspectives on the overall experience and to identify their preferred method.

As *AlignPin* method cannot complete object rotation and object-based alignment tasks, it only completed the first phase of the user study. Align-Box and *WIM* methods completed both phases of the user study. Then, we evaluated these methods based on both objective and subjective metrics.

We quantified performance using the following objective metrics, chosen to provide a comprehensive view of efficiency, physical effort, and the quality of the final alignment:

- Completion Time (T, in seconds): This is a primary measure of task efficiency.
- Hand Moving Distance (D, in meters): This metric quantifies the physical effort required from the user, a key ergonomic factor in VR interactions.

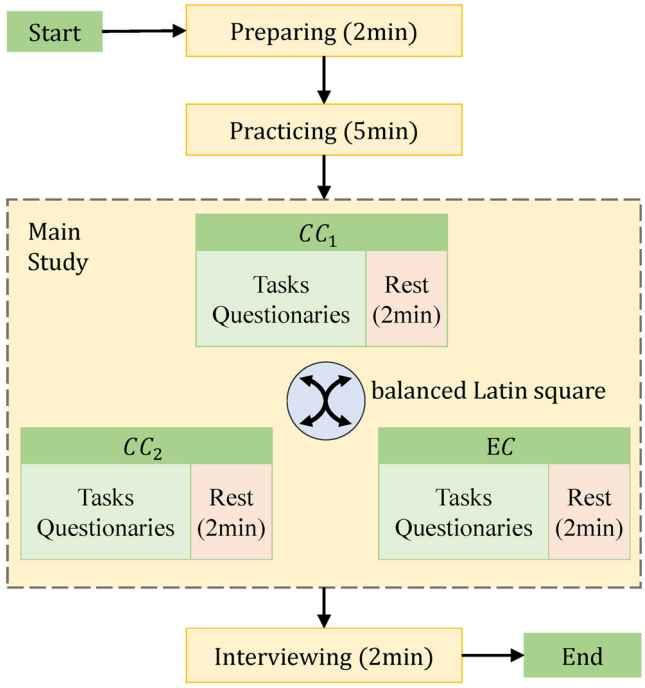

Figure 9. The procedure of the user study for each participant.

- Operation Count (N, in times): This reflects the cognitive complexity and efficiency of the interaction workflow. A lower count suggests a more streamlined technique.
- Translation Error (mm): This measures the internal geometric consistency of the alignment. After task completion, we compute a best-fit plane for the centers of the aligned objects. The error is the average of the perpendicular distances from each object's center to this plane.
- Rotation Error (degrees): This metric was evaluated in two ways depending on the task. For single-object rotation tasks (e.g., the computer monitor), it is the absolute angular difference between the object's final orientation and the target orientation. For group alignment tasks relative to a world plane (e.g., aligning flower pots to the wall), it is the angle between the normal of the objects' best-fit plane and the normal of the target plane.

Then we evaluated the user task load and usability performance with the standard NASA-TLX questionnaire (Hart, 2006; Hart & Staveland, 1988), and various aspects of usability measured with the usability questionnaire (Kim et al., 2015).

4.5. Results and discussion

We analyzed the collected data using paired-sample t-tests. A Bonferroni correction was applied for post-hoc pairwise comparisons to account for multiple comparisons. Before conducting the tests, we checked the data for normality using the Shapiro-Wilk test. In cases where the normality assumption was violated, we used the non-parametric Wilcoxon signed-rank test instead. We report exact p-values (unless $p < .001$) and 95% confidence intervals (CI) for effect sizes (Cohen's d).

4.5.1. Objective metrics

We performed a paired posthoc t-test reporting Bonferroni corrected p-values to compare the Align-Box to the control condition methods, and we quantified effect size using the expanded Cohen's d (Sawilowsky, 2009). The d values were translated to qualitative effect size estimates of Huge ($d \geq 2.0$), Very Large ($2.0 > d \geq 1.2$), Large ($1.2 > d \geq 0.8$), Medium ($0.8 > d \geq 0.5$), Small ($0.5 > d \geq 0.2$), and Very Small ($0.2 > d \geq 0.01$).

Table 5 shows the metrics results of the user study for phase $P1$. Compared to CC_1 and CC_2, the completion time is reduced by 54.4% and 77.3% by our Align-Box method. The differences are statistically significant ($p < 0.001$), and the effect sizes are "Very Large" and "Huge," respectively. As expected, the ability to quickly select target objects and group-based aligning operations reduces the completion time substantially. For the hand moving distance metric, our Align-Box method also significantly ($p < 0.001$) reduced it by 30.4% and 48.7% compared to CC_1 and CC_2.

The effect sizes are "Large" and "Very large," respectively. We believe this is because the EC method needs fewer selecting operations and does not have to operate each object separately. The operation count results corroborate this speculation. The Align-Box method shows a statistically significant ($p < 0.001$) reduction in operation count, 65.1% and 78.6% for CC_1 and CC_2. The effect sizes are "Huge." As the Align-Box simplifies the selection process, and the aligning operations apply to all selected objects, these features are important for reducing the operation count metric.

Table 6 shows the metrics results of the user study for phase $P2$. Since the CC_1 method cannot accomplish the task for the next three alignment types in Table 4. Corresponding to Phase 2, we only compared our Align-Box with the CC_2 method. For the three metrics, our Align-Box performed statistically significant ($p < 0.001$) reduction, 59.3% for completion time, 30.5% for hand moving distance, and 55.7% for operation times. The effect sizes are "Very Large," "Large," and "Huge," respectively. Combining three metrics results, our Align-Box reduced the overall participant effort, which leads to improved multiple objects alignment capability.

To assess the tradeoff between speed and precision, we also analyzed the alignment accuracy. These alignment quality metrics are summarized in Table 7.

The alignment quality results reveal two significant findings. First, for rotational precision, Align-Box was substantially more accurate than WIM, reducing the average rotation error by nearly half (from 9.8 to 5.1 degrees). This suggests that manipulating the orientation of the near-field *Proxy Box* provides more stable and precise control than orienting objects within the WIM's miniature view. As AlignPin could not perform these rotational tasks, no comparison was possible.

Second, the translation error results highlight a key difference in interaction paradigms. This metric assesses the final co-planarity of the aligned objects. For constraint-based methods like Align-Box and AlignPin, the algorithm programmatically snaps objects to a geometric plane, resulting in perfect internal consistency and a measured error of 0. In contrast, the WIM method relies on manual placement for

Table 5. Objective metrics analysis of phase P1.

Metric	Cond.	Avg ± std. dev.	Red.	p	Cohen's d	95% CI	Effect size
T	CC_1	28.42±11.72	54.4%	<0.001	1.74	[1.02, 2.45]	Very large
	CC_2	57.06±23.63	77.3%	<0.001	2.59	[1.71, 3.45]	Huge
	EC	12.94±4.50					
D	CC_1	8.79±3.46	30.4%	<0.001	0.96	[0.41, 1.50]	Large
	CC_2	11.91±3.83	48.7%	<0.001	1.91	[1.16, 2.65]	Very large
	EC	6.11±1.92					
N	CC_1	14.33±3.35	65.1%	<0.001	3.90	[2.76, 5.01]	Huge
	CC_2	23.33±6.51	78.6%	<0.001	3.90	[2.76, 5.01]	Huge
	EC	5.00±1.33					

Table 6. Objective metrics analysis of phase P2.

Metric	Cond.	Avg ± std. dev.	Red.	p	Cohen's d	95% CI	Effect size
T	CC_2	166.47±81.92	59.3%	<0.001	1.59	[0.90, 2.27]	Very large
	EC	67.78±31.72					
D	CC_2	32.53±13.42	30.5%	<0.001	0.90	[0.36, 1.43]	Large
	EC	22.61±7.80					
N	CC_2	68.06±19.55	55.7%	<0.001	2.31	[1.48, 3.12]	Huge
	EC	30.17±12.403					

Table 7. Analysis of alignment quality metrics across all tasks.

Metric	AlignPin (CC_1)	WIM (CC_2)	Align-Box (EC)
Translation error (mm)	0 ± 0	25.2 ± 6.5	0 ± 0
Rotation error (deg)	–	9.8 ± 4.2	5.1 ± 2.4

each object, leading to a significant non-zero error (25.2 mm) that reflects human imprecision. This demonstrates a core advantage of constraint-based techniques in achieving perfect geometric order.

4.5.2. Subjective metrics

Figure 10(A) shows the task load scores for all three methods. Pair-wise, the Bonferroni corrected t-test reveals that the overall task load of the Align-Box is significantly smaller than that of the AlignPin and WIM methods. For the six NASA TLX subscores, compared to the WIM method, the Align-Box performed a statistically significant reduction. Compared to the AlignPin method, there are no significant differences for Physical and Temporal subscores. For other subscores, the Align-Box method reduces them significantly.

Figure 10(B) shows the usability scores for all three methods. We also conducted a pairwise, Bonferroni-corrected t-test to analyze the data. The analysis results show that our Align-Box performs the best usability among the participants. Compared to the WIM method, our Align-Box method reveals statistically significant improvement across all six subscores. Compared to the AlignPin method, our Align-Box method significantly improves the *Intuitiveness* subscore. There are no significant differences for other subscores.

4.5.3. Discussion

The positive user study outcomes are built upon a robust technical foundation. The high gesture recognition accuracy (96.2%) and low system latency (∼48.3 ms) reported in Section 3.5 ensured a fluid interaction, allowing us to confidently attribute the observed performance gains to our interaction

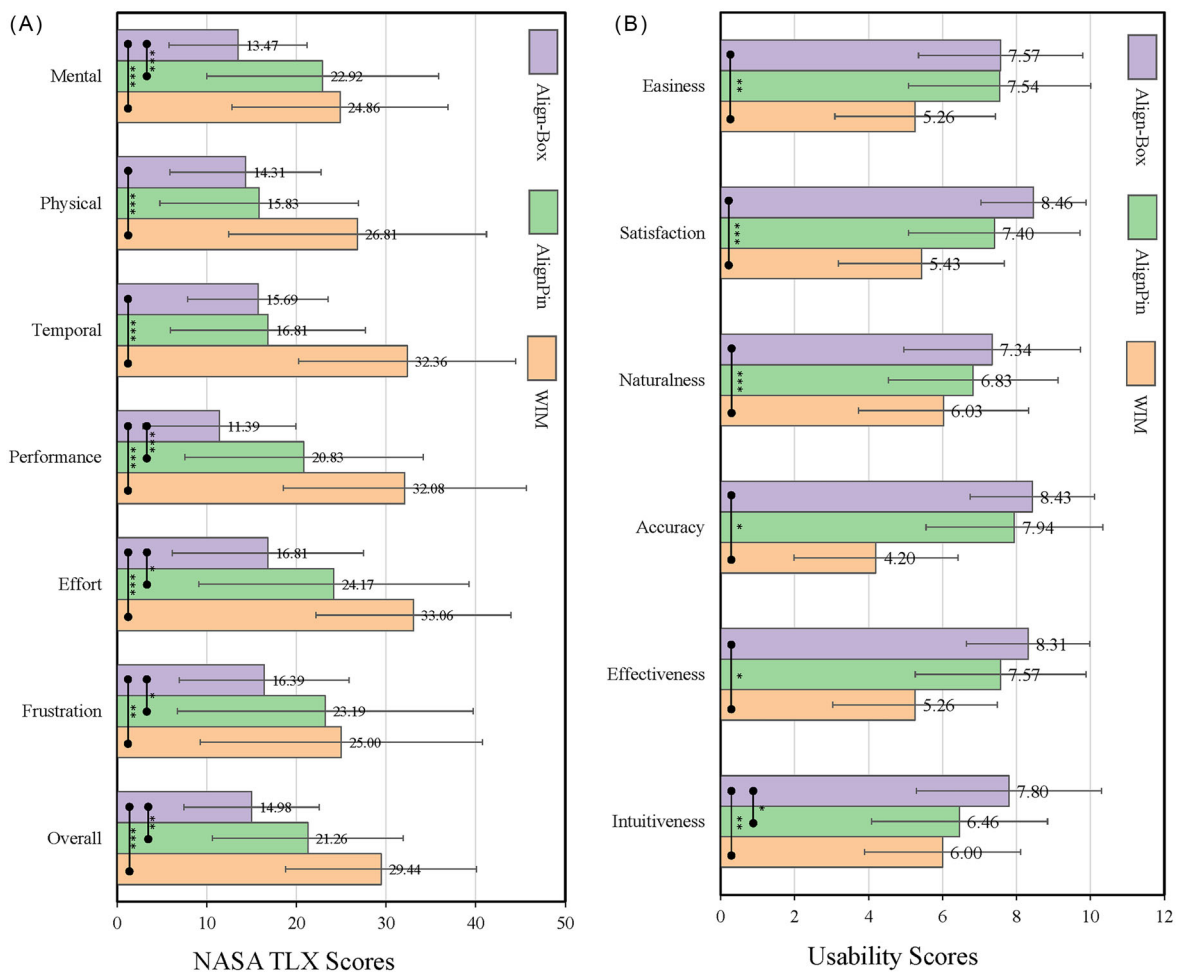

Figure 10. Subjective metrics scores of the three methods. The error bars indicate standard deviation, and statistically significant effects are marked with *, ∗∗, and ∗∗∗, representing a significance level of 0.05, 0.01, and 0.001, respectively. (A) NASA-TLX subscores and overall scores. (B) Usability subscores.

design rather than confounding system-level factors. In conclusion, our proposed Align-Box multi-object alignment method performs well in the scenarios tested.

Regarding alignment efficiency improvement, we believe our method's *Alignment Box* plays an important role. First, the *Alignment Box* allows the user to quickly batch-select objects in the scene by selecting objects at the boundaries and building the box, which saves the time and cost of selecting individual objects one by one. Second, the design saves time describing different alignment task types by quickly defining alignment types and parameters through the interrelationship between the surface of the *Alignment Box* and the objects inside and outside it. Instead of controlling the position and orientation of each object to be aligned individually, the user can consider all the objects in the *Alignment Box* as a whole, and their relationship with the reference of the alignment task can be defined and implemented through the medium of the *Alignment Box*, which also saves a lot of time. Finally, the user's operations on the *Alignment Box* are mapped through the *Proxy Box*, which is always within reach of the front of the user's body during the alignment process. The *Alignment Axis* and *Pin Button* provide the user with a convenient means of defining the alignment. As a result, our Align-Box method achieves a significant improvement in the efficiency of the alignment task. This result supports **H1**.

Regarding hand-moving distance and operation counts, we believe that the performance improvement of the Align-Box method mainly comes from the operation mechanism of the *Proxy Box* and the implementation of batch alignment. First, the design of the *Proxy Box* allows the user to operate closer to the front of the body without large hand movements. Theoretically, this benefit can be explained by Multiple Resource Theory (MRT) (Wickens, 2002). By decoupling the visual planning (observing distant objects) from the motor execution (manipulating the near-field proxy), Align-Box minimizes the conflict between visual and motor cognitive resources, reducing the need to coordinate large-scale body movements with fine-motor precision. Second, the Align-Box method is performed in batches for all selected objects in both the selection and alignment processes, significantly reducing the number of operations required. As a result, less time consumption, ease of operation, and fewer operations also reduce the user's workload (as shown in Figure 10(A)).

For usability, the Align-Box method exhibits the best usability scores (Figure 10(B)). We believe this is mainly due to the abovementioned features, such as high efficiency and low workload.

4.5.3.1. Qualitative feedback. The quantitative results were reinforced by qualitative feedback from post-study interviews. A majority of participants (32 out of 36) expressed a strong preference for Align-Box, frequently praising its speed and intuitiveness. One participant noted, "The proxy box was like a magic remote; I could arrange everything on the far table without even moving my feet." This aligns with the high Intuitiveness scores observed. Another participant praised the selection efficiency: "The multi-selection felt very fluid; I didn't have to think about aiming at each object individually." However, participants also provided valuable suggestions regarding occasional tracking loss when hands overlapped, highlighting the need for robust handling of occlusion. This result supports **H2**.

We also analyzed the influence of user experience and trial order. When stratifying users, we observed that while experienced VR users adapted faster to the bare-hand gestures, novices showed the most significant relative improvement compared to the baseline WIM method, suggesting high approachability. Regarding learning effects, while a minor general learning trend existed due to task familiarity, the counterbalancing design ensured it did not systematically skew the comparative results between methods.

4.5.3.2. Implications for Industry 5.0. Beyond basic scene editing, Align-Box holds significant potential for Industry 5.0 applications, particularly in AR/VR-assisted assembly and digital twin interactions (Eswaran et al., 2023, 2024). The ability to rapidly align virtual components to physical references using the Box-Align feature supports the "Operator 5.0" vision, empowering human workers with efficient digital tools for smart manufacturing contexts.

5. Conclusion, limitations and future work

We propose Align-Box, a bare-hand interaction metaphor for multi-object alignment in VR environments. Align-Box consists of the *Proxy Box*, *Alignment Box*, *Alignment Axis*, and *Pin Button*. It enables

fast and convenient alignment of multiple objects, both object-based and group-based. moreover, we also define a set of gestures that consists of only three types of gestures to interact with the Align-Box, making it easy to use and enhancing compatibility. Finally, we conducted an empirical user study to evaluate the performance of our method, compared with the AlignPin and WIM methods. The results show that our method is an efficient multi-object alignment tool with high usability and a low task load, for the types of tasks evaluated in our study.

5.1. Roadmap for adoption and future applications

Looking forward, we envision Align-Box as a valuable component in future VR and AR systems, particularly in content creation and industrial applications. For designers, our work offers several recommendations: maintain a minimal gesture vocabulary to reduce cognitive load and ensure the proxy interaction space is ergonomically placed. Align-Box could be integrated as a plugin for popular game engines like Unity and Unreal, enhancing in-VR level editing. In industrial workflows, such as factory floor planning or architectural visualization, our method could accelerate the arrangement of complex layouts. Its principles could also be applied in augmented reality systems for worker assistance, guiding the placement of real-world objects in assembly tasks (Marino et al., 2021; Moghaddam et al., 2021). Furthermore, the ability to rapidly configure scenes has strong implications for human-robot collaborative systems, where Align-Box could be used to define workspaces and task sequences through immersive visualization (Suzuki et al., 2022).

The reliance on optical, inside-out hand tracking introduces several limitations. First, our findings are specific to the performance of the Meta Quest 2 hardware; the generalizability of our technique's speed and comfort benefits to other VR/AR platforms with different tracking technologies remains to be investigated. Second, the system is sensitive to environmental factors such as poor lighting, which can degrade tracking quality. Third, a key negative result observed in our experiment was the occasional failure of hand tracking. While infrequent, tracking loss was most likely to occur when users held their hands close together with palms facing each other, a pose that could happen during two-handed adjustments of the *Proxy Box*. This failure case, often caused by self-occlusion, sometimes led to minor user frustration and highlights a clear limitation of the current implementation on commodity hardware. Moreover, our results are based on the Meta Quest 2; the generalizability of these findings to other VR headsets with different tracking technologies (e.g., magnetic tracking or wider camera FOVs) remains to be verified.

Furthermore, the user study was conducted within a single office scene with a moderate number of objects of similar sizes. This limited task diversity may not fully represent the scalability of Align-Box for more complex scenarios, such as very dense scenes with numerous objects, items with extreme variations in scale (very small or very large), or situations with heavily overlapping bounding boxes. While we believe the core proxy-box metaphor is scalable, further studies are needed to evaluate its performance and potential usability challenges under these more demanding conditions. However, it is worth noting that while our study focused on static objects, the core selection and alignment logic is adaptable to moving objects with minor algorithmic adjustments (e.g., increasing the bounding box update frequency and implementing predictive tracking), allowing for potential use in dynamic scenes.

A further limitation of our study is that all participants were right-handed. Our interaction design is largely symmetrical; for instance, the *Proxy Box* can be manipulated with either hand, and gestures like *Poke* and *Grab* are not specific to one hand. However, actions involving the hand cursor for initial selection were typically performed with the dominant right hand. It remains an open question whether left-handed users would perform differently. Future work should include left-handed and ambidextrous participants to validate the generalizability of Align-Box and investigate any potential performance differences related to handedness.

Finally, our user study was conducted in a single session, which does not capture long-term effects. A longitudinal study would be necessary to evaluate aspects such as skill retention over time and the onset of fatigue after prolonged, repeated use. While our initial findings on comfort are positive, a long-term evaluation is needed to fully understand the ergonomics and sustained usability of Align-Box.

One limitation of our Align-Box is that it does not consider the individual object's align property. This means that the alignment process will not change an object's relative orientation in relation to the Align-Box. For example, if some of the selected objects have non-orthogonal faces, it is hard to align them. Future work could add geometry analysis functions to recognize the faces or symmetric planes of the individual object and align them to the *Alignment Box* to enable customized alignment.

Another limitation is that the current alignment does not consider intersections between the selected objects. For example, if the projected footprint intervals of the selected objects on the XZ plane overlap, the objects will intersect with each other. Future work can investigate how to align the objects evenly or in multiple layers.

The final limitation of our method is that it primarily applies to alignment along straight lines or planes. In a broader sense, alignment should refer to the arrangement of objects, including the alignment along planes or lines discussed in this article, as well as alignment along various geometric shapes. Future work could explore how to achieve more personalized alignment of multiple objects in virtual scenes or how to implement esthetically pleasing or semantically based alignment through scene perception. Additionally, future work will focus on integration with adaptive AR guidance or mixed-reality alignment frameworks to further enhance its industrial applicability.

Future work will also involve more detailed follow-up analyses. A post-hoc analysis of the interaction logs could reveal finer-grained user strategies and common error patterns. Furthermore, dedicated studies are needed to explore the relationship between object size and alignment precision, as well as to formally quantify the impact of partial hand occlusion on gesture recognition performance. These analyses will help to further refine the interaction design and better characterize the technique's performance envelope.

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Author contributions

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Appendix A. Nomenclature

Table A1. List of abbreviations and key terms.

Term/abbreviation	Description
VR	Virtual reality
Align-Box	The proposed multi-object alignment interaction metaphor.
Proxy Box	The near-field, interactable proxy of the Alignment Box.
Alignment Box	The bounding box in the scene that encloses all selected objects.
Pin Button	An interactive button on an Alignment Axis used to trigger alignment.
Alignment Axis	An axis on the Proxy Box used for rotation and alignment control.
CC_1, CC_2	Control condition 1 (AlignPin) and 2 (WIM).
EC	Experimental condition (Align-Box method).
T, *D*, *N*	Objective metrics: Completion time, hand moving distance, and operation count.
IPD	Interpupillary distance.
IRB	Institutional review board.

Appendix B. Geometric calculations

To streamline the methodology section, we present the detailed geometric calculations for the Align-Box proxy mechanism here.

B1. Normal vector calculation

For the object-based alignment, the target normal vector $\overrightarrow{n}_{target}$ is computed from the selected surface points P_1, P_2, P_3 as:

$$\overrightarrow{n}_{target} = \frac{(P_2 - P_1) \times (P_3 - P_1)}{||(P_2 - P_1) \times (P_3 - P_1)||} \tag{B1}$$

The alignment rotation matrix R_{align} is then derived to map the current box normal $\overrightarrow{n}_{box}$ to $\overrightarrow{n}_{target}$.

B2. Proxy projection matrix

The mapping between the *Alignment Box* (B_A) and the *Proxy Box* (B_P) relies on a scaling factor S and a translation vector $\overrightarrow{t}$. The projection of a hand interaction point p_{hand} to the scene object space p_{scene} is defined as:

$$p_{scene} = S^{-1} \cdot R_{sys}^T \cdot (p_{hand} - C_{proxy}) + C_{align} \tag{B2}$$

where C_{proxy} and C_{align} are the centroids of the Proxy Box and Alignment Box, respectively, and R_{sys} represents the current system rotation.